Search, Matching, Turnover, and Unemployment

Labor II — Week 4

Teaching deck draft

Central question and course repositioning

- Once workers and firms must search for each other, how are hiring, turnover, and unemployment jointly determined?
- Week 4 is the first major market-interaction week in Labor II.
- The focus is worker flows, vacancies, job ladders, and unemployment dynamics rather than internal firm organization alone.

Week 3 to Week 4 bridge

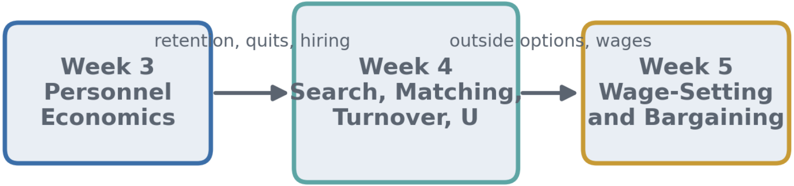
- Week 3 studied incentives, promotions, retention, and internal labor markets inside the firm.
- Week 4 asks how quits, layoffs, recruitment, and mobility aggregate into labor-market flows across firms and states.
- Week 5 will add wage posting, bargaining, and wage-setting once outside options are fully in view.

Why search frictions matter

- Workers do not meet vacancies costlessly.
- Jobs end, new jobs open, and many moves happen directly from job to job.
- Unemployment is a flow equilibrium outcome, not a static leftover stock.

Course map

Labor II pivots from the inside of the firm to market-wide labor allocation



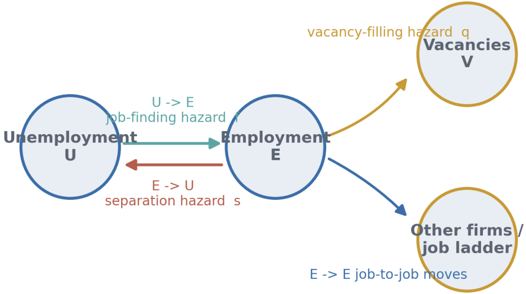
Matching function and worker-side versus vacancy-side hazards

$$M_t = \mathcal{M}(U_t, V_t), \qquad f_t = \frac{M_t}{U_t}, \qquad q_t = \frac{M_t}{V_t}$$

- The same meeting technology implies different hazards for workers and firms.
- f_t is a job-finding rate; q_t is a vacancy-filling rate.
- Matching is a reduced-form meeting object, not yet a full wage or sorting theory.

Worker flows and hazards

Week 4 tracks three transition margins, not only unemployment exits



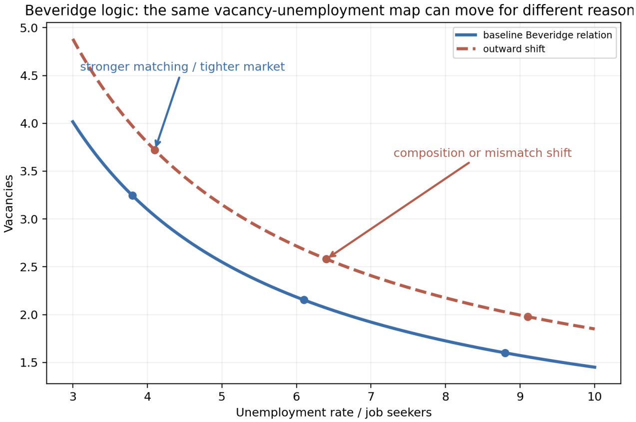
Flow unemployment and steady-state accounting

$$u_{t+1} - u_t = s_t(1 - u_t) - f_t u_t$$

$$u^* = \frac{s}{s + f}$$

- Unemployment depends jointly on inflows from employment and outflows to jobs.
- A high-unemployment market can reflect high separations, low job finding, or both.
- A theory of unemployment without separations is incomplete.

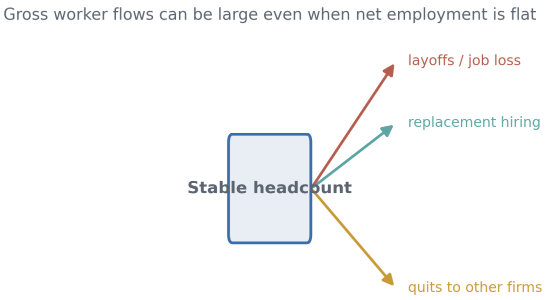
Beveridge logic and matching efficiency

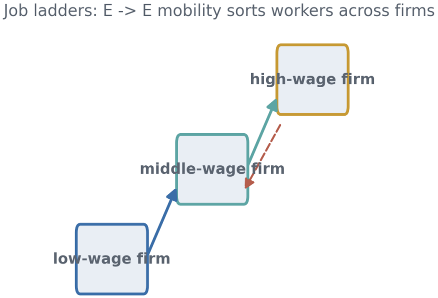


Separations, turnover, layoffs, job destruction, and churning

- A separation ends a worker-firm spell; a layoff is one type of separation.
- Job destruction is a job-flow object, not a synonym for worker turnover.
- Churning is worker reallocation beyond what net employment change requires.
- Stable headcount can hide intense replacement hiring and restructuring.

Separations, churning, and job ladders





Job-to-job transitions, on-the-job search, and job ladders

$$\lambda_{it}^{EE} = \lambda\big(w_{it}, x_i, \phi_t, \text{firm quality}_j\big)$$

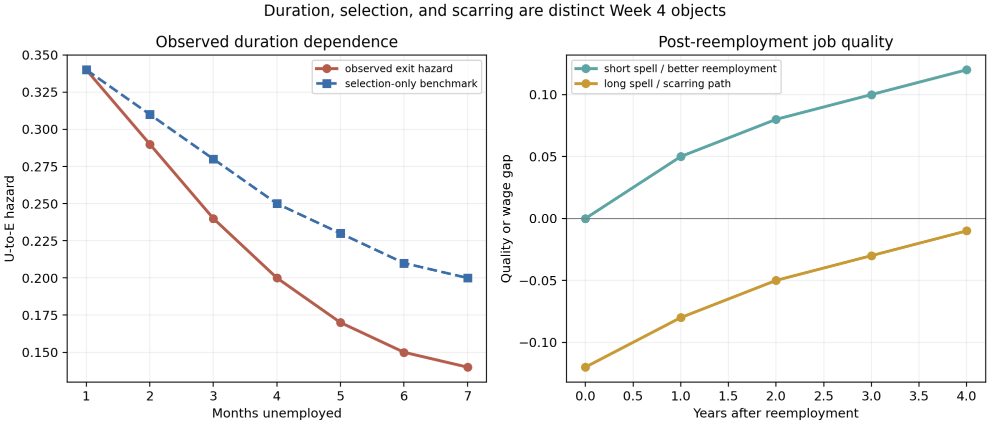
- Many labor-market transitions never pass through unemployment.
- Employed search drives poaching, retention pressure, wage growth, and sorting.
- Job ladders flatten in weak markets even before all adjustment shows up in unemployment.

Unemployment duration, search intensity, and unemployment effects

$$y_i^{\text{reemp}} = \alpha + \beta D_i + \gamma X_i + \varepsilon_i$$

- Observed duration dependence is not automatically a causal unemployment effect.
- Search intensity, beliefs, employer screening, and dynamic selection can all matter.
- The post-unemployment object includes job quality and scarring, not just exit hazards.

Duration dependence and scarring



Empirical designs and what they identify

Design	Variation	Margin	Hardest unobserved object
CPS flows	Worker-month transitions	U-E, E-U	Match quality, recruiting intensity
Vacancy evidence	Time or place vacancies	Vacancy filling, Beveridge shifts	Searcher composition, mismatch
Linked worker-firm data	Worker moves across firms	E-E, churning, ladders	Opportunity set, reasons for moves
Search-platform data	Applications over spell	Search intensity, targeting	Representativeness
Field experiments or policy shocks	Randomized histories, UI rules	Duration, callbacks, job quality	General-equilibrium spillovers

Frontier extension

- Beliefs and expectations: perceived hazards shape search effort.
- Digital search traces: applications reveal targeting and intensity, but coverage is incomplete.
- Selective hiring and mismatch: Beveridge shifts can reflect composition, not just efficiency.
- Scarring and firm quality loss: recession timing changes later opportunities and job ladders.

Bridge to Week 5 wage-setting

- Once vacancies take time to fill and workers have outside options, the wage is no longer an anonymous competitive parameter.
- Search frictions create the environment for wage posting, bargaining, retention pressure, and rent sharing.
- Week 5 asks how wages are set once this Week 4 mobility structure is in place.

Takeaways

- Search theory is about worker flows, vacancies, separations, mobility, and unemployment jointly.
- The key distinctions are U-E versus E-U versus E-E, worker flows versus job flows, and matching efficiency versus composition.
- Modern evidence combines transition data, vacancy data, linked worker-firm records, digital search traces, and policy or field variation.
- Week 5 turns these mobility and outside-option objects into wage-setting theory.